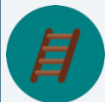


1.5 Transformations on the Plane – Part 1

Lesson Introduction

 Benchmark	MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.	 Learning Targets	- I can identify the angle of rotation of a transformation of a figure on a plane. - I can identify the center of rotation of a transformation of a figure on a plane. - I can identify the direction of a rotation of a transformation of a figure on a plane. - I can distinguish a rotation from other transformations.		
 Benchmark Coherence	Building On		Working Toward		
	MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.		N/A		
 Essential Questions	- How can the center of rotation be found? - Is rotation a rigid transformation? - Is there more than one way to define the rotation of a preimage to image?				
 Lesson Narrative	Throughout this lesson, students observe and measure rotations using different methods. By creating a rotation, reflection, and translation of the same preimage, comparisons between the differences of each transformation are made through the lens of an Error Analysis. Students engage in determining the direction, center, and degree of rotation of multiple transformations.				
 Component Summary	1.5.1 Warm-Up (~10 min.)	Creating images of different views (SE p. 28)	1.5.3 Guided Practice (5-10 min.)	Describing rotations (SE p. 32)	
	1.5.2 Guided Instruction (25 min.)	Describe transformations on plane (SE p. 29)	1.5.4 Wrap-Up (~5 min.)	Describing rotations (SE p. 33)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Glossary:</u> - Clockwise - Counterclockwise - Degree		- Patty Paper - Ruler - Protractor (Optional) Markers or Colored Pencils		For 1.5.2 Guided Instruction, consider providing a copy on patty paper of the preimage and point of rotation for students who struggle with tracing precisely.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse clockwise and counterclockwise. - Students may enter this lesson with the idea that a reflection is the same as a 180° rotation. - Students may incorrectly use the protractor to measure the angles.	N/A
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share There are a variety of opportunities for students to Think-Pair-Share throughout 1.5.2 Guided Instruction. Students need the opportunity to think through each of the discussion questions, process the information in pairs, and share their results to the class.	MA.912.MTR.5.1: Patterns and Structure Students relate what they previously learned in grade 8 to further understand rotations. Students have to create plans and logically order steps to determine the center of rotation, the degree of rotation, and the direction of rotation. Students will look for similarities amongst the rotations to help understand why many rotations can be described in more than one way.
 Differentiate Instruction	Students have an opportunity to physically engage with the measurement of angles in 1.5.2 Guided Instruction. This is an opportunity to encourage students to play with the different angles and possibly test each other on angles. Students could draw angles and ask partners to find the measure of the angles.	
Lesson Notes:		

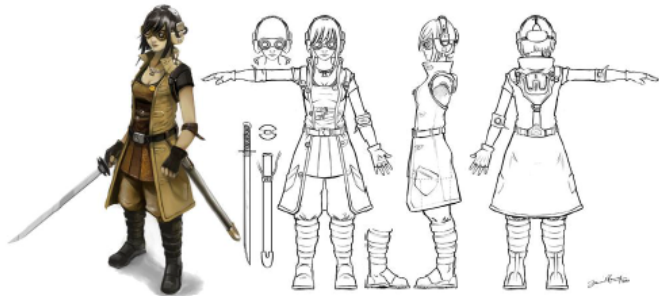
1.5.1 Warm-Up: Rotations in Drawings

Component Overview: Students consider how images will be drawn and seen from different views.



1.5.1 Warm-Up: Rotations in Drawings

The picture shows the front, side, and back view of an animated character.



1. What do you notice about all three views — front, side, and back, of the animated character?
Answers will vary. I notice that it is the same person, just turned around.

When animators and cartoonists sketch the movement of a character, they use a sheet like the one shown.

Front	Side	Back



Students are encouraged to provide any answer they can think of. The way students see this variety of perspectives could help students understand the idea of rotation, which will be the focus of the lesson.

1.5.1 Warm-Up: Rotations in Drawings (continued)

2. Use one of the pictures below, or your own idea, to sketch the three views. Use the dotted lines to help you align key features of the picture.



Answers will vary.

Front	Side	Back



Consider speaking with students about how rotations of a character depicts movement in a three-dimensional space.



Students may struggle with the idea of having to draw. Encourage students to try so that they develop an understanding of rotations. Remind students that this is an opportunity to develop a growth mindset.

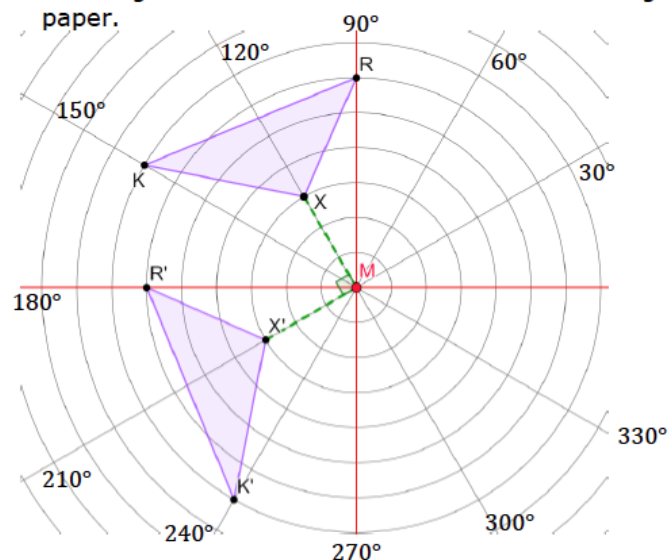
1.5.2 Guided Instruction: Transformations on the Plane

Component Overview: Students determine the angle of rotation between a preimage and an image using circular grid paper and with a protractor. Students draw and compare the image of a rotation to the image of a reflection and a translation of the same preimage.



1.5.2 Guided Instruction: Transformations on the Plane

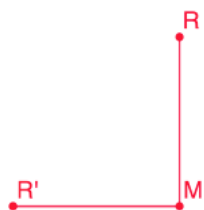
1. The diagram shows $\triangle R K X$ and $\triangle R' K' X'$ on circular grid paper.



Follow the steps below.

- Use a piece of patty paper to copy $\triangle R K X$ and point M .
- Place the patty paper so that it aligns with $\triangle R K X$ and point M .
- Place the point of your writing utensil on point M and rotate $\triangle R K X$ so that it lays on top of $\triangle R' K' X'$.

- a. Draw $\angle R M R'$.



Students often struggle with rotations. By introducing the circular grid paper students can relate the angle measures on the grid to the rotation. It may be helpful to ask students what they notice and wonder about the diagram. Help students notice that the vertices of the preimage and the image are on the same circle. Students can count the number of degrees from preimage vertex to the image vertex or notice that the angle of the vertex X of the preimage, the center of rotation, and vertex X of the image create a right angle.

1.5.2 Guided Instruction: Transformations on the Plane (continued)

- b. Draw $\angle KMK'$.



- c. Complete the table.

Angle	Measurement (degrees)
$\angle XMX'$	90°
$\angle RMR'$	90°
$\angle KMK'$	90°

- d. Discuss with your partner how the angle measures on the circular grid can be used to verify the rotation and how to determine the direction of rotation. Summarize your discussion.

Answers will vary. Subtract the degree measure of the new point minus the degree measure of the original point. The direction depends on if the degree measure increased (counterclockwise) or decreased (clockwise) when moving from the original point to the new point.

- e. Identify the transformation of $\triangle RKX$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is <u>M</u>. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	<i>The transformation was a 90° rotation counterclockwise around point M.</i>



Students may ask for a protractor here to measure the angles or may provide an estimation. Direct students back to the circular graph to determine the degree measures.



Think-Pair-Share

For question 1d, students could be instructed to begin working independently with about 30 seconds of “think time.” After this, students could work with their partner to discuss the ideas they developed. The pair of students should develop a summary that could be shared with the class.

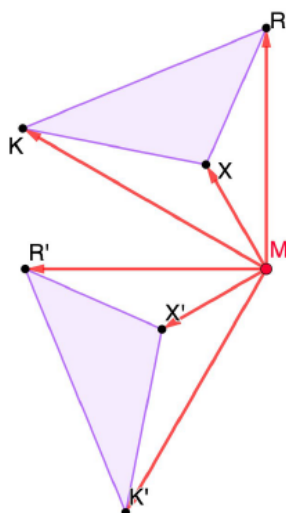
1.5.2 Guided Instruction: Transformations on the Plane (continued)

- f. Discuss with your partner whether or not the transformation could be described as "a rotation of 270° clockwise centered at point M ." Explain why or why not.

Yes, moving in the other direction would be "a rotation of 270° clockwise centered at point M " because all of the vertices of triangle RKX will map onto triangle $R'K'X'$..

2. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ without the circular grid paper.

- a. Draw $\angle XMX'$, $\angle RMR'$, and $\angle KMK'$.



- b. Discuss with your partner how the degree of rotation can be determined using the angles that connect the vertices and the center of rotation.

Answers will vary.



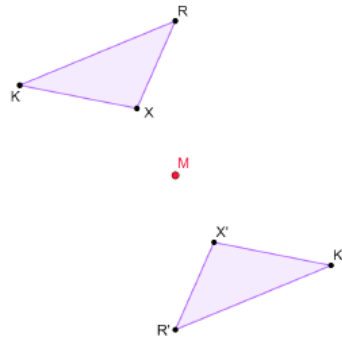
Spend time on question 1f to ensure students understand why rotations can be described in two different ways.



Question 2 is the same set of triangles and rotation as question 1. This helps students build their understanding of rotations without any grid. Help students recognize that the degree, center, and direction of the rotation can be determined without the circular grid.

1.5.2 Guided Instruction: Transformations on the Plane (continued)

3. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ using a different transformation.



- a. Identify the transformation of $\triangle RKX$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
☐ Dilation ☒ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☒ Center of rotation is <u>M</u>. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	<i>The transformation was a 180° rotation counterclockwise around point M.</i> <i>The transformation was a 180° rotation clockwise around point M.</i>

- b. Two students' descriptions of the transformation of $\triangle RKX$ are shown.

Michela	Marc
a reflection across line M	a translation down

- Use a piece of patty paper to copy $\triangle RKX$.
- Perform each student's transformation to prove that each transformation does not work.



Students are given a rotation of 180° . Help students relate that the angle is 180° , because there is a straight line through the center of rotation from each point on the preimage to the image.



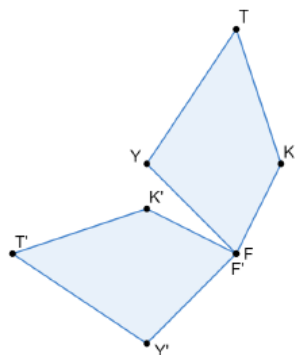
Consider having a student perform the reflection and the translation to demonstrate Michela's and Marc's transformations. Alternately, you can show each using software such as GeoGebra.

1.5.2 Guided Instruction: Transformations on the Plane (continued)

- c. Discuss with your partner why both students are wrong. Write your reasoning for why the transformations do not work.

Answers will vary. Michela is incorrect because her answer would have points R' and K' on opposite sides of X' . Marc is also incorrect because the point X' is below $R'K'$ after his transformation instead of above it.

- d. The diagram shows $TKFY$ and $T'K'F'Y'$.



Identify the transformation of $TKFY$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is F. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	<i>This is a 90° rotation Counterclockwise about point F.</i>



Students may struggle understanding why the reflection is not correct. Specifically point out that the vertices do not align.



For a visual entry-point, consider having students use colored markers to draw the angles from preimage point to image point. Then, using a protractor, students can measure the degree of rotation.



When giving descriptions of the answers always include the alternate rotation.

1.5.3 Guided Practice: Rotations

Component Overview: Students determine the direction of rotation, center of rotation, and degree of rotation for three different transformations.



1.5.3 Guided Practice: Rotations

The table shows three different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

Transformation	Direction, Center, and Degree
	- Direction of rotation is <u>Clockwise</u>. - Center of rotation is <u>P</u>. - Degree of rotation is <u>90°</u>.
	- Direction of rotation is <u>Clockwise</u>. - Center of rotation is <u>T</u>. - Degree of rotation is <u>90°</u>.
	- Direction of rotation is <u>Clockwise</u>. - Center of rotation is <u>P</u>. - Degree of rotation is <u>180°</u>.



To add complexity, students can determine the degree of rotation for each transformation if rotated both clockwise and counterclockwise.

1.5.4 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a Ticket Out the Door by identifying the direction, center, and degree of each rotation.



1.5.4 Wrap-Up: Ticket Out the Door

1. The table shows two different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

Transformation	Direction, Center, and Degree
	- Direction of rotation is Counterclockwise. - Center of rotation is K. - Degree of rotation is 180°.
	- Direction of rotation is Counterclockwise. - Center of rotation is P. - Degree of rotation is 90°.



Use student understanding to inform your instruction and to determine where in future lessons and units you can help students build a better understanding of rotations. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.